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Sound producing device and method

Abstract

The sound producing device includes a shutter element configured to obscure an aperture positioned to receive an amplitude-modulated acoustic wave; a blind element, wherein the aperture is formed within the blind element; and an amplitude-modulation means, configured to generate an amplitude-modulated acoustic wave toward the aperture; wherein the amplitude-modulated acoustic wave is produced according to an input audio signal.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8861752	12/2013	Margalit	381/431	G10K 15/04
9774959	12/2016	Kupershmidt	N/A	G10K 15/04
9866948	12/2017	Margalit	N/A	H04R 17/00
9913048	12/2017	Margalit	N/A	H04R 19/02
10034098	12/2017	Kupershmidt	N/A	B81B 3/0021
10123126	12/2017	Margalit	N/A	H04R 1/24
10284961	12/2018	Margalit	N/A	H04R 19/005
10779088	12/2019	Liang	N/A	H03K 5/1565
11172310	12/2020	Liang	N/A	H04R 1/025
11343609	12/2021	Margalit	N/A	H04R 19/02
12107546	12/2023	Liang	N/A	H03D 7/165
2016/0360321	12/2015	Margalit	N/A	H04R 19/02
2017/0064457	12/2016	Kupershmidt	N/A	H04R 19/02
2022/0183659	12/2021	Margalit	N/A	A61B 8/5207
2023/0247357	12/2022	Margalit	381/98	H04R 19/02

OTHER PUBLICATIONS

Carl Poldy, Headphones Fundamentals Tutorial AES 120, May 2006. cited by examiner

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/540,937, filed on Sep. 28, 2023. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present application relates to a sound producing device and a sound producing method, and more particularly, to a sound producing device and a sound producing method free from harmonic distortion and inter-modulation distortion.

2. Description of the Prior Art

(2) Loudspeaker design has changed little in nearly a century. A loudspeaker (or “speaker”) is an electro-acoustic transducer that produces sound in response to an electrical signal input. The electrical signal causes a vibration of the speaker cone in relation to the electrical signal amplitude.

(3) Speaker may operate in ultrasonic frequency. Previous ultrasonic speakers suffer from harmonic distortion and inter-modulation distortion, which degrades sound quality.

(4) Therefore, it is necessary to improve the prior art.

SUMMARY OF THE INVENTION

(5) It is therefore a primary objective of the present application to provide a sound producing device and method, to improve over disadvantages of the prior art.

(6) An embodiment of the present application discloses a sound producing device. The sound producing device comprises a shutter element configured to obscure an aperture positioned to receive an amplitude-modulated acoustic wave; a blind element, wherein the aperture is formed within the blind element; and an amplitude-modulation means, configured to generate an amplitude-modulated acoustic wave toward the aperture; wherein the amplitude-modulated acoustic wave is produced according to an input audio signal.

(7) An embodiment of the present application discloses a sound producing method applied for a sound producing device. The sound producing method comprises applying a first control signal on an amplitude-modulation means, such that the amplitude-modulation means produces an amplitude-modulated acoustic wave toward an aperture, wherein the first control signal is generated according to an input audio signal; applying a second control signal on a shutter element or a blind element, wherein the second control signal is irrelevant to the input audio signal; wherein the sound producing device comprises the shutter element, the blind element and the amplitude-modulation means; wherein the aperture is formed within the blind element; wherein the shutter element obscures the aperture positioned to receive the amplitude-modulated acoustic wave.

(8) An embodiment of the present application discloses a sound producing method applied for a sound producing device. The sound producing method comprises an amplitude-modulation means producing an amplitude-modulated acoustic wave toward an aperture; and a shutter element or a blind element performing a compressed-and-expanded movement; wherein the sound producing device comprises the shutter element (**201**), the blind element (**102**) and the

amplitude-modulation means; wherein the aperture is formed within the blind element (102); wherein the shutter element (201) obscures the aperture (120) positioned to receive the amplitude-modulated acoustic wave.

(9) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of a sound producing device (SPD) in the art.
- (2) FIG. 2 is a schematic diagram of an SPD according to an embodiment of the present application.
- (3) FIG. 3 illustrates spectral density of an acoustic wave produced by the SPD shown in FIG. 1.
- (4) FIG. 4 illustrates spectral density of an acoustic wave produced by the SPD shown in FIG. 2.
- (5) FIG. 5 is a schematic diagram of an SPD according to an embodiment of the present application.

DETAILED DESCRIPTION

(6) FIG. 1 illustrates a schematic diagram of a sound producing device (SPD) 10 in the art, similar to one disclosed in U.S. Pat. No. 9,913,048. The SPD 10 comprises a shutter element 101, a blind element 102 and an oscillation element 103. The shutter element 101, the blind element 102 and the oscillation element 103 may be silicon membrane fabricated using MEMS technology. The oscillation element 103 receives a control signal $v_{sub.13}$ so as to generate a constant-amplitude acoustic wave $P_{sub.1}$, as ultrasonic carrier signal, toward a direction orthogonal to a surface of the oscillation element 103. In an embodiment, the constant-amplitude acoustic wave $P_{sub.1}$ may be expressed as $P_{sub.1} = P_{sub.0} \cdot \sin(2\pi f_{sub.c} t + \phi_{sub.1})$, where $P_{sub.0}$ is constant, $f_{sub.c}$ is ultrasonic carrier frequency and $\phi_{sub.1}$ represents phase of $P_{sub.1}$. The constant amplitude wave $P_{sub.1}$ passes through an aperture 120 formed within the blind element 102 and through a gap/passage 122 between the shutter element 101 and the blind element 102, and radiates outward as an acoustic wave 140. A chamber may be formed between the oscillation element 103 and the blind element 102, and walls of the chamber are omitted for brevity.

(7) The shutter element 101 (or the blind element 102) receives a control signal $v_{sub.11}$ and vibrates accordingly. In an embodiment, the gap distance $d_{sub.1}$ may be expressed as $d_{sub.1} = d_{sub.0} \cdot \sin(2\pi f_{sub.c} t + \phi_{sub.2})$, where $x(t)$ is an audible source signal or an input audio signal within an audible band, $\phi_{sub.2}$ represent a phase, $d_{sub.0}$ is a natural spacing between shutter element 101 and blind element 102 when neither of them is driven, $A_{sub.x}$ is a parameter (related to modulation depth) to control the dynamic range so that $|A_{sub.x} \cdot \sin(2\pi f_{sub.c} t + \phi_{sub.2})| < 1$.

(8) For the SPD 10, the control signal v_i is generated according to the input audio signal $x(t)$. Suppose displacement of the shutter element 101 is proportional to the control signal $v_{sub.11}$, the control signal $v_{sub.11}$ may comprise a component related to $x(t) \cdot \sin(2\pi f_{sub.c} t + \phi_{sub.2})$. On the contrary, in order to generate the constant-amplitude acoustic wave $P_{sub.1}$, the control signal $v_{sub.13}$ may comprise a component related to $\sin(2\pi f_{sub.c} t + \phi_{sub.1})$. Note that, the control signal $v_{sub.13}$ comprises no component which is related to $x(t)$.

(9) The $x(t)$ -relevant control signal $v_{sub.11}$ and the $x(t)$ -irrelevant control signal $v_{sub.13}$ may be generated according to a control signal generator 160. The shutter element 101 and the oscillation element 103 may be coupled to the control signal generator 160.

(10) By controlling the gap distance $d_{sub.1}$ between the shutter element 101 and the blind element 102 in a time varying manner, an output acoustic wave 140 may be a kind of amplitude modulated acoustic wave. Specifically, the gap 122 would introduce an acoustic resistance $R_{sub.1}$ and the output acoustic wave 140 would have an output volume velocity $U_{sub.1}$ as $U_{sub.1} = P_{sub.1} / R_{sub.1}$. Since $R_{sub.1}$ is related to $d_{sub.1}$ and $d_{sub.1}$ comprises component related to $x(t)$, $U_{sub.1}$ comprises component related to $x(t)$. Hence, the SPD 10 is able to produce sound related to $x(t)$.

(11) However, the construct of the SPD 10 brings harmonic distortion and inter-modulation distortion. It is mainly because the $U_{sub.1}$ comprises cubic term of $x(t)$, which is nonlinear with respect to $x(t)$.

(12) According to Carl Poldy, Tutorial AES 120, Appendix A.3, Paris, May 2006, the acoustic resistance of the acoustic passage 122 is approximately proportional to a reciprocal of a cube of $d_{sub.1}$, i.e., $R_{sub.1} \propto 1/(d_{sub.1})^3$. The output volume velocity $U_{sub.1}$ would be

$$U_{sub.1} = P_{sub.1} / R_{sub.1} \propto P_{sub.0} \cdot \sin(2\pi f_{sub.c} t + \phi_{sub.1}) \cdot d_{sub.1}^3 \quad (\text{eq. 2.1})$$
$$= P_{sub.0} \cdot \sin(2\pi f_{sub.c} t + \phi_{sub.1}) \cdot d_{sub.0}^3 [1 + A_{sub.x} \cdot \sin(2\pi f_{sub.c} t + \phi_{sub.2})]^3 \quad (\text{eq. 2.2})$$

(13) Note that, a cubic term of $x(t)$ is embedded inside a component of $U_{sub.1}$. Illustratively, assuming $\phi_{sub.1} = \phi_{sub.2} = 0$, $U_{sub.1}$ can be expressed as $U_{sub.1} = U_{sub.0} \sin \theta [1 + A_{sub.x} x(t) \sin \theta]^3$ (eq. 3), where θ represents $2\pi f_{sub.c} t$, i.e., $\theta = 2\pi f_{sub.c} t$, and $U_{sub.0}$ represents a peak level of the output volume velocity wave $U_{sub.1}$ when none of the elements 101, 102, 103 is driven. By applying a demodulation operation and/or applying a filtering operation to capture component within an audible spectrum band, a resulting component $y(t)$, which may be expressed as (eq. 4), shall be obtained. Note that, (eq. 4) is derived by expanding right-hand-side of (eq. 2.2) and thus (eq. 5) is obtained. Note that, the signal component of $U_{sub.1}/U_{sub.0}$ within (eq. 5) which would fall/lie within the audible

spectrum band is to terms in the 2.sup.nd square bracket within the curly brace of (eq. 5), where a cubic term of x(t) is included therein.

$$(14) \quad y(t) = \text{audible_band_filtered}[\frac{U_1}{U_0}] = \frac{3}{2}A_x x(t) + \frac{3}{8}A_x^3 x^3(t). \quad (\text{eq. 4})$$

$$\frac{U_1}{U_0} = \sin n \{ [1 + \frac{3}{2}A_x^2 x^2(t)] + [3A_x x(t) + \frac{3}{4}A_x^3 x^3(t)] \sin + [-\frac{3}{2}A_x^2 x^2(t)] \cos 2 + [-\frac{1}{4}A_x^3 x^3(t)] \sin 3 \}. \quad (\text{eq. 5})$$

(15) One of the root causes of significant total distortion in the sound produced by the device in SPD **10** is that the audible source signal x(t) shows up as a factor in the control signal that modifies acoustic resistance related to the acoustic output nonlinearly. One remedy to lower the distortion in the reproduced sound is to avoid including the audible band source in the control signal that will alter an acoustic property (e.g., acoustic resistance) nonlinearly.

(16) FIG. **2** illustrates a schematic diagram of an SPD **20** according to an embodiment of the present application. SPD **20** is similar to SPD **10**, and thus, same components are denoted by same notations. The SPD **20** comprises a shutter element **201**, the blind element **102** and an oscillation element **203**. Instead of generating the constant-amplitude acoustic wave P.sub.1, the oscillation element **203**, driven by a control signal v.sub.23, is configured to generate an amplitude-modulated acoustic wave P.sub.2 toward the aperture **120** or toward a direction orthogonal to a surface of the oscillation element **203** (which may be regarded as a modulation operation). That is, an amplitude of the acoustic wave P.sub.2 is corresponding to the input audio signal x(t). The shutter element **201**, driven by a control signal v.sub.21 and configured to obscure the aperture **120** positioned to receive the amplitude-modulated acoustic wave P.sub.2, may perform a constant amplitude or x(t)-irrelevant oscillation/vibration (which may be regarded as a demodulation operation), such that an acoustic wave **240** is produced and radiates outward.

(17) The x(t)-irrelevant control signal v.sub.21 and the x(t)-relevant control signal v.sub.23 may be generated according to a control signal generator **260**. The shutter element **201** and the oscillation element **203** may be coupled to the control signal generator **260**.

(18) The amplitude-modulated acoustic wave P.sub.2 may be regarded as double sideband with suppressed carrier (DSB-SC) modulated (see (eq. 6) below). In an embodiment, driving circuit disclosed in U.S. application Ser. No. 18/665,525 may be included in the control signal generator **260**, especially for generating the control signal v.sub.23, but not limited thereto.

(19) In an embodiment, the control signal v.sub.23 may drive the oscillation element **203** such that the amplitude-modulated acoustic wave P.sub.2, as illustrated in FIG. **2**. In an embodiment, P.sub.2 may be expressed as P.sub.2=P.sub.0 x(t) sin(2πf.sub.ct) (eq. 6) (neglecting phase for brevity), which is x(t)-relevant. On the other hand, the control signal v.sub.21 may drive the shutter element **201** such that a gap distance d.sub.2 corresponding to a gap **222** between the shutter element **201** and the blind element **102** may be expressed as d.sub.2=d.sub.0.Math. [1+A.sub.d.Math.sin(2πf.sub.ct)] (eq. 7) (neglecting phase for brevity), which is x(t)-irrelevant, where parameter Aa parameterizes the maximum displacement of the shutter element **201** with |A.sub.d|<1. The acoustic resistance R.sub.2 corresponding to the gap **222** or the gap distance d.sub.2 may be expressed as (eq. 8). Given that, the output volume velocity U.sub.2 may be expressed as (eq. 9).

(20)

$$R_2 = R_0 [1 + A_d \sin(2 f_c t)]^{-3} = R_0 \{ [1 + \frac{3A_d^2}{2}] + [3A_d + \frac{3A_d^3}{4}] \sin(2 f_c t) + [-\frac{3A_d^2}{2}] \cos(2 f_c t) + [-\frac{A_d^3}{4}] \sin(2 f_c t) \}^{-1} \quad (\text{eq. 8})$$

$$\frac{U_2}{U_0} = x(t) \sin(2 f_c t) \{ [1 + \frac{3A_d^2}{2}] + [3A_d + \frac{3A_d^3}{4}] \sin(2 f_c t) + [-\frac{3A_d^2}{2}] \cos(2 f_c t) + [-\frac{A_d^3}{4}] \sin(2 f_c t) \} \quad (\text{eq. 9})$$

(21) In (eq. 8), no x(t) term is shown therein. In (eq. 9), those four terms in the curly brackets on the right-hand-side may be seen as four processes in parallel to demodulate the amplitude-modulated signal x(t) sin(2πf.sub.ct). The 1.sup.st term is simply a gain of 1+3A.sub.d.sup.2/2, and it produces an output within an ultrasonic band [f.sub.c-BW, f.sub.c+BW] that is above audible frequencies. Herein BW denotes audible spectrum bandwidth, e.g., BW=20 KHz or 24 KHz for high resolution audio application, but not limited thereto. The 2.sup.nd term is to demodulate x(t) sin(2πf.sub.ct) into two parts: one within the band [0, BW] that is audible and proportional to x(t), and the other within an ultrasonic band [2f.sub.c-BW, 2f.sub.c+BW] that is not/beyond perceivable. The 3.sup.rd term converts x(t) sin(2πf.sub.ct) into two parts, one within an ultrasonic band [f.sub.c-BW, f.sub.c+BW] and the other within another ultrasonic band [3f.sub.c-BW, 3f.sub.c+BW]. Neither of them in the 3rd term is perceivable. The 4th term converts x(t) sin(2πf.sub.ct) into two parts, one within an ultrasonic band [2f.sub.c-BW, 2f.sub.c+BW] and the other within another ultrasonic band [4f.sub.c-BW, 4f.sub.c+BW]. Neither part in the 4th term is perceivable.

(22) Note that, only the term of [3A.sub.d+3A.sub.d.sup.3/4] in the 2.sup.nd square bracket within the curly brace of U.sub.2/U.sub.0 in of (eq. 9) would be related to component within the audible spectrum band, and no cubic term of x(t) is included in the [3A.sub.d+3A.sub.d.sup.3/4] term. The resulting component y(t) is therefore derived as (eq. 10), which is linear with respect to x(t), the input audio signal.

$$(23) \quad y(t) = \text{audible_band_filtered}[\frac{U_2}{U_0}] = [\frac{3}{2}A_d + \frac{3A_d^3}{8}]x(t) \quad (\text{eq. 10})$$

(24) As can be seen from (eq. 10), resulting component y(t) is linear with respect to the input audio signal x(t), and thus total harmonic distortion (THD) and total distortion (TD) no longer exist, analytically.

(25) Numerically, please refer to FIG. **3** and FIG. **4** (as a glimpse). FIG. **3** illustrates (normalized) spectral density of the

acoustic wave **140** produced by SPD **10**, and FIG. **4** illustrates (normalized) spectral density of acoustic wave **240** produced by SPD **20**. In FIG. **3** and FIG. **4**, the dotted lines indicate audible bandwidth BW as 24 KHz. $x(t)$ is set to be a 4-tone signal at {310, 800, 2000, 3200} Hz, $A_{sub.x}=0.4$ for FIG. **3** and $A_{sub.d}=0.4$ for FIG. **4** are employed in the simulation environment.

(26) It can be also seen from FIG. **3**, the acoustic wave **140** would have inter-modulated components/interference which would fall/lie within the audible spectrum band, and thus total distortion increases. On the other hand, the acoustic wave **240**, produced by SPD **20**, has no THD and TD. Note that, the noise floors showing up in FIG. **4** with levels below roughly -270 dB arise from numerical errors of digital computer.

(27) Furthermore, TABLE I tabulates the 3^{sup}.rd harmonic distortion $HD_{sub.3}$ and total distortion TD for various $x(t)$, where $x(t)$ may be single tone or multi-tone signal, with $A_{sub.x}=0.4$ and $A_{sub.x}=0.9$ for the SPD **10**.

(28) TABLE-US-00001 TABLE I $A_{sub.x} = 0.4$ $A_{sub.x} = 0.9$ $x(t)$ $HD_{sub.3}$ TD $HD_{sub.3}$ TD 1-tone 0.97% 0.97% 4.40% 4.40% 1000 Hz (-40.26 dB) (-40.26 dB) (-27.14 dB) (-27.14 dB) 2-tone 0.24% 1.07% 1.14% 4.95% {700, 1100} (-52.23 dB) (-39.45 dB) (-38.89 dB) (-26.10 dB) Hz 3-tone 0.45% 1.13% 2.10% 5.26% {700, 1000, (-46.96 B) (-38.97 dB) (-33.58 dB) (-25.59 dB) 1300} Hz 4-tone 0.22% 1.01% 1.06% 4.83% {310, 800, (-53.07 dB) (-39.91 dB) (-39.49 dB) (-26.33 dB) 2000, 3200} Hz 8-tone 0.11% 0.58% 0.54% 2.84% {60, 116, (-59.16 dB) (-44.79 dB) (-45.32 dB) (-30.95 dB) 224, 432, 834, 1610, 3108, 6000} Hz

(29) $HD_{sub.3}$ and TD may be elaborated as follow. In general, the tones in the output are classified in three groups. Group 1 represents fundamental frequencies, which exist in the input signal. Group 2 represents harmonic frequencies, which are multiple of any of the fundamental frequencies. These tones in Group 2 account for the harmonic distortion. Group 3 represents any frequencies other than the fundamental or harmonic frequencies. These tones in Group 3 account for the inter-modulation distortion.

(30) As quality of sound produced by a speaker/SPD may be accessed through performance metrics such as harmonic distortions ($HD_{sub.n}$ and THD) and total distortion (TD), $HD_{sub.n}$, THD and TD may be defined as follows:

$$(31) HD_n (n\text{-th Harmonic Distortion}) = \frac{RMS\{ \text{.Math. (all } n\text{-th harmonic of fundamental frequencies)} \}}{RMS\{ \text{.Math. (all components in Group 1)} \}}$$

$$THD (Total Harmonic Distortion) = \frac{RMS\{ \text{.Math. (All components in Group 2)} \}}{RMS\{ \text{.Math. (All components in Group 1)} \}}$$

$$TD (Total Distortion) = \frac{RMS\{ \text{.Math. (All components in Group 2 and 3)} \}}{RMS\{ \text{.Math. (All components in Group 1)} \}}$$

(32) It should be noted that all the summations in the above definitions are carried over audible frequency range, i.e., $[0, BW]$, e.g., $[0 \text{ Hz}, 24 \text{ kHz}]$. RMS represents root mean square operation.

(33) Several observations are obtained via numerical simulations corresponding to SPD **10**. As the number of tones in the source signal increases, the number of inter-modulated components grows rapidly in the output acoustic wave. In addition, as $A_{sub.x}$ increases, the levels of the fundamental components in the output acoustics increase nonlinearly and the levels of the 3rd-harmonic components and the inter-modulated components increase as well.

(34) Above are THD and TD results for SPD **10**. As for SPD **20**, under same simulation scenario, there are no harmonic components or inter-modulated components within the audible band of the resulting component $y(t)$ produced by SPD **20**. That is, SPD **20** produces the acoustic wave **240** with $THD=TD=0$. Quality of sound of SPD **20** significantly outperforms which of SPD **10**.

(35) Note that, the $x(t)$ -irrelevant control signal $v_{sub.21}$ applied to the shutter element (e.g., **201**) is merely for illustration purpose, which is not limited thereto. For example, the $x(t)$ -irrelevant control signal $v_{sub.21}$ may be applied to the blind element (e.g., **102**) as well. As long as (the gap (e.g., **222**) between) the shutter and blind elements perform a compressed-and-expanded movement at an ultrasonic carrier frequency (e.g., $f_{sub.c}$), e.g., an amplitude or a time-variant function of the gap distance is irrelevant to $x(t)$ such as the one shown as (eq. 7), requirements of the present application is satisfied, which is within the scope of the present application.

(36) Furthermore, using the (MEMS-fabricated) oscillation element (e.g., **203**) to produce the amplitude-modulated acoustic wave $P_{sub.2}$ is merely for illustration purpose, which is not limited thereto. For example, any kind of amplitude-modulation means or amplitude modulator (e.g., **303** shown in FIG. **5**) capable of generating amplitude-modulated acoustic wave (modulated according to $x(t)$ such as the one shown as (eq. 6)) may be employed within the SPD of the present application. The amplitude-modulation means or amplitude modulator **303** may comprise (MEMS-fabricated) oscillation element, diaphragm, film, coil, membrane, piston, etc. As long as the amplitude-modulation means or amplitude modulator can generate amplitude-modulated acoustic wave, requirements of the present application is satisfied, which is within the scope of the present application.

(37) By avoiding the audible band source in the control signal that will alter acoustic property (e.g., acoustic resistance) nonlinearly, in the present application, the audio sound is produced by generating amplitude-modulated acoustic wave according to the audible band source (input audio signal) and performing compressed-and-expanded movement which is irrelevant to the audible band source. Analytical and numerical results demonstrate not only input-output linearity but also harmonic distortion and inter-modulation distortion vanish. Therefore, sound quality of SPD of the present application would be significantly enhanced.

(38) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A sound producing device, comprising: a shutter element, configured to obscure an aperture positioned to receive an amplitude-modulated acoustic wave; a blind element, wherein the aperture is formed within the blind element; and an amplitude-modulation means, configured to generate an amplitude-modulated acoustic wave toward the aperture; wherein the amplitude-modulated acoustic wave is produced according to an input audio signal; wherein the shutter element or the blind element receives a first control signal; wherein the first control signal is irrelevant to the input audio signal.
 2. The sound producing device of claim 1, wherein the amplitude-modulation means receives a second control signal; wherein the second control signal is generated according to the input audio signal, such that the amplitude-modulated acoustic wave is produced according to the input audio signal.
 3. The sound producing device of claim 1, wherein the shutter element or the blind element is driven by the first control signal, such that a gap between the shutter element and the blind element is compressed-and-expanded at an ultrasonic carrier frequency of the amplitude-modulated acoustic wave.
 4. The sound producing device of claim 1, wherein a time-variant function of a gap distance of the gap between the shutter element and the blind element is irrelevant to the input audio signal.
 5. The sound producing device of claim 1, wherein the amplitude-modulation means comprises an oscillation element; wherein the oscillation element produces the amplitude-modulated acoustic wave toward a direction orthogonal to a surface of the oscillation element.
 6. The sound producing device of claim 1, wherein the shutter element or the blind element is driven by the first control signal which is irrelevant to the input audio signal.
 7. A sound producing method, applied for a sound producing device, the sound producing method comprising: applying a first control signal on an amplitude-modulation means, such that the amplitude-modulation means produces an amplitude-modulated acoustic wave toward an aperture, wherein the first control signal is generated according to an input audio signal; applying a second control signal on a shutter element or a blind element, wherein the second control signal is irrelevant to the input audio signal; wherein the sound producing device comprises the shutter element, the blind element and the amplitude-modulation means; wherein the aperture is formed within the blind element; wherein the shutter element obscures the aperture positioned to receive the amplitude-modulated acoustic wave.
 8. A sound producing method, applied for a sound producing device, the sound producing method comprising: an amplitude-modulation means producing an amplitude-modulated acoustic wave toward an aperture; and a shutter element or a blind element performing a compressed-and-expanded movement; wherein the sound producing device comprises the shutter element, the blind element and the amplitude-modulation means; wherein the aperture is formed within the blind element; wherein the shutter element obscures the aperture positioned to receive the amplitude-modulated acoustic wave; wherein the shutter element or the blind element receives a control signal; wherein the control signal is irrelevant to the input audio signal.
 9. The sound producing method of claim 8, comprising: producing, by the amplitude-modulation means, the amplitude-modulated acoustic wave according to an input audio signal at an ultrasonic frequency; and the shutter element or the blind element performing the compressed-and-expanded movement at the ultrasonic frequency; wherein the compressed-and-expanded movement is irrelevant to the input audio signal.
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